Table 3. Quality and serving cost of three attention variants. Hybrid is best on every metric at 16k context; Dense runs out of memory at batch 32.

		16k context			32k
Metric	**Unit**	**Dense**	**Sparse**	**Hybrid**	**Hybrid**
Quality · 1,000 requests					
Task success	% ↑	74.2	74.3	**76.6**	78.0
Calibration error	pp ↓	4.8	4.2	**3.5**	3.2
Serving · batch 4					
P95 latency	ms ↓	184.2	157.9	**149.8**	230.3
Throughput	req/s ↑	30.4	34.7	**36.2**	23.5
Peak memory	GB ↓	36.4	30.1	**28.5**	44.8
Energy / request	J ↓	9.8	8.2	**7.6**	12.4
Stress · batch 32					
P95 latency	ms ↓	*OOM*	392	**351**	*OOM*
Throughput	req/s ↑	*OOM*	61.3	**66.9**	*OOM*

Synthetic data. One 80 GB GPU, BF16 weights, 256 output tokens per request. Bold: best of the three 16k variants; the 32k column is context and is not ranked. OOM = out of memory (an outcome, not a zero).